

# Mitigating Media Manipulation : DeepFake Detection using Deep Neural Networks

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**Abstract** – In today's digital landscape, the emergence of deepfake technology poses a significant risk, enabling the creation of highly convincing counterfeit videos through rapid advancements in computational power. Our research addresses this challenge by developing a deep learning-based method to detect deepfake videos. Leveraging artificial intelligence, we integrate the pre-trained Xception model for feature extraction and Inception for temporal sequence analysis to meticulously analyze video frames for subtle manipulation indicators. Additionally, we strategically freeze certain layers of the Xception model to maintain accuracy while conserving computational resources. Through extensive evaluations on diverse datasets like Face-Forensic++, Deepfake Detection Challenge, and Celeb-DF, we demonstrate the reliability of our approach in distinguishing authentic from deepfake videos across various scenarios. By prioritizing simplicity and robustness, our method offers a practical solution to combat the pervasive threat of artificially generated fake videos, contributing to mitigating societal risks associated with the proliferation of deepfake technology and safeguarding against the spread of misinformation and manipulation in an increasingly digitized world.

**Keywords:** *DeepFake detection, CNN, GAN, RNN, LSTM.*

## I. INTRODUCTION

In recent times, the collaboration between Artificial Intelligence (AI) and deep learning techniques has led to significant progress across diverse domains, particularly in computer vision. Deep learning models have gained the ability to analyze and classify a wide range of parameters, encompassing human faces to intricate medical images. Although the inception of deep learning dates back to the late 1990s, its practical application surged in the early 2020s, driven by enhanced computational capabilities available at reduced costs and the unprecedented availability of data facilitated by the proliferation of the internet.

Despite the myriad benefits associated with artificial intelligence, it's imperative to acknowledge the accompanying drawbacks of this cutting-edge technology. With every advancement in technology, there emerges a new set of challenges. Among these challenges, one of the most recent and menacing is the proliferation of deep fake technology. While the term "deep fake" itself may not inherently imply negativity, its implications and potential consequences are concerning.[2]

The advent of artificial intelligence (AI) has ushered in remarkable progress, but it has also introduced new challenges, prominently among them being the rise of deep fake technology. Deepfake videos, manipulated

using Machine Learning-based algorithms to swap individuals in existing images or videos, have become a pressing concern in modern society. These videos, now one of the paramount challenges facing civilization, are frequently employed to defame and tarnish the reputations of celebrities, politicians, and other notable figures, thereby wreaking havoc within communities and across nations.

These manipulated visuals typically involve overlaying a target person's face onto a source video. There are two main types of deep fakes: those with cloned voices and those without. Deep fakes with cloned voices enhance the authenticity of the content by combining fake visuals with synthesized voices generated from a target person's voice and source audio. This manipulation makes it appear as if the target person is speaking words they never actually said.[2]

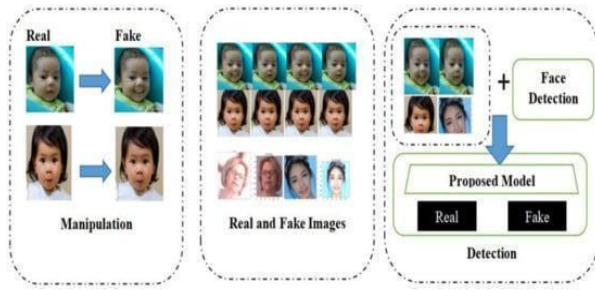


Figure 1. A high-level view of the model for creating and detecting deep fakes[3].

The exponential growth of AI and accessible deep learning-based tools, such as convolutional neural networks, autoencoders, generative adversarial networks, and platforms like TensorFlow, Keras, and PyTorch, has democratized image and video editing, rendering it accessible to almost anyone with a computer. Deep fakes, which can include manipulated voices, present a significant threat as they blur the distinction between reality and fabrication. By seamlessly integrating cloned voices into source audio, deep fakes become increasingly difficult to detect, heightening the risk of misinformation and fraudulent activities. The accessibility of deep fake creation tools further exacerbates the issue, emphasizing the urgent need for proactive measures in detection and mitigation. Through heightened awareness and effective strategies, we can work towards mitigating

the detrimental effects of digital manipulation and fortify defenses against the proliferation of deceptive content.

## II. LITERATURE SURVEY

The proliferation of deepfake videos underscores the urgency in devising robust detection mechanisms. Recent research endeavors have harnessed advancements in both hardware and software technologies to confront this pressing issue. Within the realm of deepfake detection, a variety of sophisticated methodologies leveraging deep learning techniques have emerged, reflecting the ongoing pursuit of effective countermeasures against deceptive content.

### 1. ResNext50 with LSTM Architecture:

In a recent study, [2] proposed a novel deepfake detection system that integrates cutting-edge hardware components, including Nvidia GeForce RTX 30 Series GPUs, for enhanced performance. This system employs CUDA 11.3 and PyTorch, a powerful machine learning framework developed by Meta AI, to harness GPU capabilities efficiently. By leveraging ResNext50 for feature extraction, LSTM models with GAN technology for classification and evaluation using a confusion matrix to measure accuracy.

The model's architecture is illustrated in detail, emphasizing the utilization of ResNext50's parallel stacking and LSTM's classification capabilities. Dropout probability and latent dimensions are specified for LSTM layers, and the model's output is processed through linear and adaptive average pooling layers before final classification via softmax.

Through rigorous testing on unseen data from the FF++ dataset, the system achieves an impressive accuracy of 97.25% for detecting deepfake videos, particularly notable for videos with specific compression levels.

### 2. Facial Image Cropping and Deepfake Detection: InceptionResNetV2, MobileNet, and DenseNet121:

[3] presents a comprehensive methodology for detecting deepfake videos, a critical challenge in contemporary media manipulation. Initially, the study

employed facial image cropping techniques, leveraging the dlib module in Python to extract facial images frame by frame from the training dataset. Subsequently, three prominent convolutional neural network (CNN) architectures, namely InceptionResNetV2, MobileNet, and DenseNet121, were employed for deepfake detection.

To enhance the models' performance, pre-trained weights from ImageNet were utilized for the initial layers, followed by a GlobalAveragePooling2D layer and a dense layer with softmax activation. The models were then compiled using the Adam optimizer with carefully chosen hyperparameters, including a learning rate of  $1e-5$  and zero decay. Evaluation metrics included the binary cross-entropy loss function and the accuracy matrix.

The research conducted training, validation, and testing using the Kaggle Deepfake Detection Challenge (DFDC) dataset, partitioning it into 70% for training, 20% for validation, and 10% for testing purposes. Additionally, results were cross-validated with a modified version of the FaceForensics++ dataset to ensure robustness and comparability.

For result analysis, the study adopted a systematic approach, extracting fixed-length images from videos to feed into the trained models. Confusion matrices were generated to visualize and interpret the models' performance. Evaluation metrics such as accuracy, precision, and recall were computed, revealing notable accuracies of 93.7%, 94.93%, and 93.86% for InceptionResNetV2, MobileNet, and DenseNet121, respectively, on the DFDC dataset.

### 3. Combination of ResNet50 and LSTM:

[4] explores several strategies for detecting deepfake videos and carefully analyzes their effectiveness through detailed result analysis.

**Residual Neural Network (ResNet-50):** This method employs a sophisticated neural network, ResNet-50, which is adept at extracting features from videos and classifying them as real or fake. By training on vast datasets of both genuine and manipulated videos,

ResNet-50 can learn to discern subtle differences between authentic and fake content.

**Pooling Layers:** Utilizing pooling layers, specifically average pooling, helps in filtering out irrelevant information from video frames. This process contributes to enhancing the accuracy of the detection model by focusing on the most relevant features.

**Long Short-Term Memory (LSTM):** LSTM, a type of recurrent neural network, is employed to analyze the temporal consistency of video frames. By considering the sequential nature of video data, LSTM can effectively identify patterns and deviations characteristic of deepfake videos.

**Hybrid Architecture:** The research proposes a hybrid approach that combines the strengths of ResNet50 and LSTM. While ResNet50 extracts high-level features from individual frames, LSTM models the temporal dependencies between these frames. This integrated approach aims to improve detection accuracy, especially for videos created using sophisticated techniques like Generative Adversarial Networks (GANs).

Result analysis demonstrates the efficacy of the proposed methodologies. Testing the model for 20 and 40 epochs yields accuracy rates of 84.75% and 87.48%, respectively. The validation and testing accuracy improve with increasing epoch count, as observed from the generated graphs. Additionally, confusion matrices aid in assessing the model's performance in testing scenario

## III. PROPOSED SOLUTION

Our methodology delves into leveraging Deep Neural Networks (DNNs) as a potent tool for detecting DeepFake content. With a focus on countering the challenges posed by AI-generated manipulated media, our approach emphasizes innovative and resilient solutions. We explore the nuanced process of utilizing DNNs to discern these intricate forgeries, offering a glimpse into the strategies, techniques, and data-centric

approaches essential for successful DeepFake detection. This methodology is pivotal in preserving digital trust and authenticity amidst the era of advanced AI technologies.

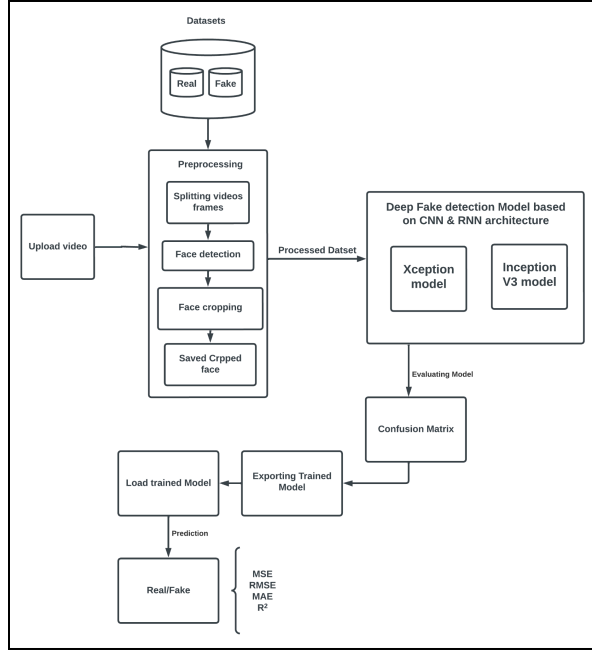


Figure 1. Workflow for detection of Deep Fake videos

### Building a Robust Foundation Dataset :

**Dataset Compilation:** We curate a diverse dataset comprising videos sourced from reputable platforms, including YouTube, FaceForensics++, and the Deep Fake Detection Challenge (DFDC) dataset. This dataset amalgamates 50% original videos with an equal proportion of manipulated deepfake videos, ensuring a balanced representation of authentic and synthetic content.

**Dataset Splitting:** The compiled dataset undergoes partitioning into distinct subsets for training, validation, and testing. Specifically, 70% of the data is allocated to the training set, 10% to the validation set, and the remaining 20% to the testing set. This partitioning strategy facilitates effective model training and unbiased evaluation.

**FaceForensics++ (FF++) Assessment:** The FaceForensics++ dataset plays a pivotal role in evaluating the performance of deep fake detection algorithms. It encompasses a spectrum of manipulation techniques, including but not limited to face swapping, reenactment, and facial expression alteration. By leveraging FF++, we assess the robustness of our model against diverse manipulation scenarios, thereby enhancing its detection capabilities.

**DFDC Benchmarking:** The Deep Fake Detection Challenge (DFDC) dataset serves as a crucial benchmark for benchmarking deep fake detection models. This dataset features meticulously crafted deep fake content generated using state-of-the-art AI techniques. Through rigorous benchmarking against DFDC, we aim to gauge our model's efficacy in detecting sophisticated manipulations encountered in real-world scenarios.

**Extracting the Essence Features :** The deep learning model employed in this study is a custom convolutional neural network (CNN), designed to classify the input as either fake or real. The architecture of the CNN consists of several layers, including convolutional, pooling, flatten, and fully connected layers. The model architecture is summarized as follows:

- 1) **Input Layer:** Accepts input images/frames with dimensions of 224x224x3 (height, width, and color channels).
- 2) **Convolutional Layers:** Utilize 3x3 kernel filters with ReLU activation function and He normal initialization. The initial convolutional layer comprises 64 filters followed by max-pooling, while subsequent layers contain 128 filters each.
- 3) **Pooling Layers:** Max-pooling layers are interspersed between convolutional layers to

downsample feature maps and extract essential information.

- 4) **Flatten Layer:** Flattens the output of the convolutional layers into a 1D vector, preparing it for input to the fully connected layers.
- 5) **Fully Connected Layers:** Consist of densely connected neural units with ReLU activation and He normal initialization. Dropout layers with a dropout rate of 0.5 are incorporated to prevent overfitting.
- 6) **Output Layer:** Comprises a single neuron with sigmoid activation, producing a binary classification output indicating the probability of an image being fake or real.

7)

```
[22]: model.compile(loss="binary_crossentropy", optimizer="nadam",
          metrics=["accuracy"])
      model.summary()
```

Model: "sequential\_2"

| Layer (type)                   | Output Shape          | Param #    |
|--------------------------------|-----------------------|------------|
| conv2d_6 (Conv2D)              | (None, 224, 224, 64)  | 9,472      |
| max_pooling2d_4 (MaxPooling2D) | (None, 112, 112, 64)  | 0          |
| conv2d_7 (Conv2D)              | (None, 112, 112, 128) | 73,856     |
| conv2d_8 (Conv2D)              | (None, 112, 112, 128) | 147,584    |
| max_pooling2d_5 (MaxPooling2D) | (None, 56, 56, 128)   | 0          |
| flatten_2 (Flatten)            | (None, 481408)        | 0          |
| dense_6 (Dense)                | (None, 128)           | 51,380,352 |
| dropout_4 (Dropout)            | (None, 128)           | 0          |
| dense_7 (Dense)                | (None, 64)            | 8,256      |
| dropout_5 (Dropout)            | (None, 64)            | 0          |
| dense_8 (Dense)                | (None, 1)             | 65         |

Total params: 51,619,585 (196.91 MB)  
 Trainable params: 51,619,585 (196.91 MB)  
 Non-trainable params: 0 (0.00 B)

Table 2 Layers of Models

**The Preprocessing Pipeline :** In the preprocessing stage of deepfake detection, our methodology focuses on optimizing computational resources while ensuring robust model training and experimentation. This section outlines our approach to handling the computational complexities associated with video processing and describes the functions and processes involved in preparing the input data for our deepfake detection model.

**Computational Efficiency Strategy :** Considering the resource-intensive nature of processing videos,

particularly a 10-second video at 30 frames per second resulting in 300 frames, we adopt a pragmatic approach. We advocate for processing only the initial 100 frames, or fewer frames as needed, to optimize computational efficiency without compromising on the quality of model training and experimentation.

**Preprocessing Functions Overview:** Our preprocessing pipeline comprises several key functions designed to extract features from videos and prepare them for input into our deepfake detection model. These functions are:  
*load\_video Function:* Reads frames from a specified video file path. Crops frame to a square shape for uniformity. Resizes frames to a predefined size (IMG\_SIZE) for consistency. Reorders color channels to ensure standardized input format.

*build\_feature\_extractor() Function :* Initializes an InceptionV3 neural network pre-trained on ImageNet for feature extraction. Constructs a model that accepts preprocessed frames as input and outputs extracted features.

*prepare\_all\_videos() Function:* Processes each video in the dataset to extract features and masks. Loads frames from videos and extracts features using the InceptionV3-based feature extractor. Generates masks to denote valid time steps within each video. Stores features, masks, and corresponding labels (0 for real videos, 1 for deepfake videos) in arrays for each video.

### Data Standardization and Model Input Preparation

: The preprocessing stage ensures that input data for our deepfake detection model is standardized and ready for further processing. This includes ensuring uniformity in input frames, efficient feature extraction, and proper labeling of data to facilitate accurate model training and evaluation. The custom CNN is tailored



for binary classification of real and fake images from video data, while the Xception model leverages pre learned features for improved detection accuracy. These methods, implemented independently, offer flexibility and enable comparative analysis for optimal deepfake detection strategies.

#### **Model Initialization with Xception Architecture:**

The Xception model, renowned for its depth and efficiency in feature extraction, was chosen as the base architecture for our deepfake detection model. By initializing the Xception model with pre-trained weights from ImageNet, we harnessed the learned features from a diverse set of images. This initialization step is crucial as it provides a starting point with rich representations of visual features, which are transferable and beneficial for various computer vision tasks.

#### **Appending Global Average Pooling and Output**

**Layers:** Following the base Xception model, we appended a global average pooling layer. This pooling layer aggregates spatial information across the feature maps, reducing the dimensionality and capturing the most salient features. Subsequently, a dense layer with a sigmoid activation function was added as the output layer. This output layer facilitated binary classification, distinguishing between real and fake videos, a fundamental aspect of deepfake detection.

**Freezing Base Model Weights:** To leverage the valuable information encoded in the pre-trained Xception model while avoiding catastrophic forgetting, we froze the weights of the base model. Freezing the weights prevents them from being updated during the initial training phase, ensuring that the model retains the general features learned during pre-training.

**Initial Training Phase:** The model underwent an initial training phase where the base model weights were kept fixed. During this phase, we employed the Stochastic Gradient Descent (SGD) optimizer with a relatively high learning rate of 0.1 and a momentum of 0.9. The binary cross-entropy loss function was used as the optimization objective, guiding the model to minimize the difference between predicted and actual labels.

**Validation and Overfitting Prevention:** Validation was an integral part of the training process to monitor the model's performance and prevent overfitting. We used a separate validation set to evaluate the model's generalization ability. Regular validation checks during training allowed us to adjust hyperparameters and detect signs of overfitting, ensuring that the model performed well on unseen data.

**Fine-Tuning Phase:** Following the initial training, we proceeded with a fine-tuning phase to adapt the model to the specific task of deepfake detection. In this phase, we made the top layers of the base Xception model trainable again, specifically unfreezing layers beyond index 56. This selective unfreezing allowed the model to learn task-specific features related to deepfake characteristics. To facilitate fine-tuning and prevent abrupt weight updates that could destabilize the model, we reduced the learning rate to 0.01. The lower learning rate ensured smoother adjustments to the weights, especially in the newly added top layers and the unfrozen layers of the base model.

#### **Continued Training and Performance Improvement**

**:** The model underwent further training for additional epochs during the fine-tuning phase. This extended training period allowed the model to learn intricate patterns and nuances specific to deepfake content.

Through continued training, the model iteratively improved its performance.

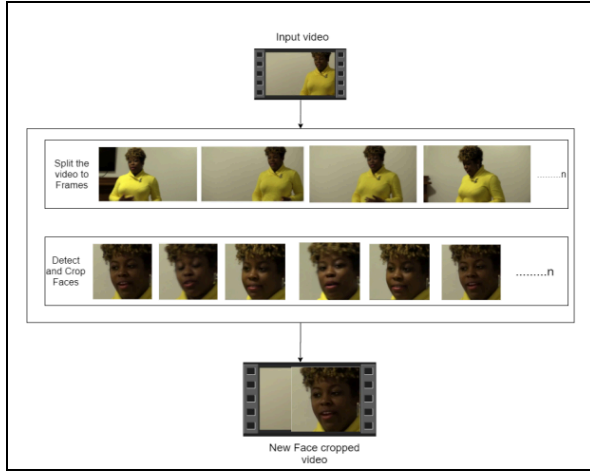


Figure 2 Pre Processing Steps

**Results :** The model outputs whether the video is a deepfake or real, along with its confidence level. This is achieved through preprocessing, frame splitting, and face cropping, directly feeding cropped frames into the model for detection. The performance of the fine-tuned model was evaluated on the test set to assess its accuracy and generalization capabilities. The model's effectiveness in detecting deepfake videos was measured in terms of binary classification accuracy and loss. Evaluation metrics were computed using the evaluate method provided by TensorFlow, allowing for an objective assessment of the model's performance.

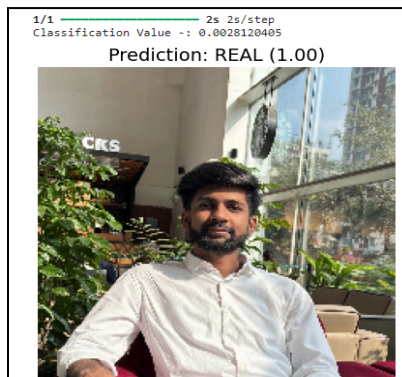


Figure 3 Prediction Result

## IV. CONCLUSION

In conclusion, our endeavor to combat the proliferation of deepfake content has yielded promising results through the implementation of diverse neural network architectures. The utilization of a custom CNN model facilitated detection of manipulated images, while the fine-tuned Xception pretrained model demonstrated proficiency in discerning subtle anomalies indicative of deepfake manipulation. Additionally, the integration of an InceptionV3-based model for video analysis showcased a comprehensive approach, effectively detecting a spectrum of potential manipulations, from traditional facial replacement to sophisticated reenactment techniques.

Notably, our method achieves commendable accuracy by analyzing just one second of video, processing 10 frames per second. The implementation incorporates a pre-trained ResNext CNN model for frame-level feature extraction and utilizes InceptionV3 for temporal sequence analysis, effectively identifying changes between consecutive frames. Initially, the pre-trained model yielded an accuracy of 60%, but through strategic training techniques, such as freezing certain layers during the initial epochs and subsequently unfreezing all layers for further refinement over 10 epochs, we achieved a notable increase in accuracy, reaching 80.8%. This significant improvement underscores the effectiveness of our approach in enhancing model performance over successive training iterations.

The entire implementation was conducted using TensorFlow and Keras libraries, providing a robust and efficient framework for developing and training deep learning models for deepfake detection. This implementation provided valuable insights into the

effectiveness of custom CNN architectures versus pretrained models for deepfake detection, paving the way for future research and development in this critical area of study.

## V. REFERENCES

- [1] H. Chotaliya, M. A. Khatri, S. Kanojiya and M. Bivalkar, "Review: DeepFake Detection Techniques using Deep Neural Networks (DNN)," 2023 6th International Conference on Advances in Science and Technology (ICAST), Mumbai, India, 2023, pp. 480-484, doi: 10.1109/ICAST59062.2023.10454938.
- [2] L. S and K. Sooda, "DeepFake Detection Through Key Video Frame Extraction using GAN," 2022 International Conference on Automation, Computing and Renewable Systems (ICACRS), Pudukkottai, India, 2022, pp. 859-863, doi: 10.1109/ICACRS55517.2022.10029095.
- [3] A. Rahman et al., "Short And Low Resolution Deepfake Video Detection Using CNN," 2022 IEEE 10th Region 10 Humanitarian Technology Conference (R10-HTC), Hyderabad, India, 2022, pp. 259-264, doi: 10.1109/R10-HTC54060.2022.9929719.
- [4] S. R. B. R, P. Kumar Pareek, B. S and G. G, "Deepfake Video Detection System Using Deep Neural Networks," 2023 IEEE International Conference on Integrated Circuits and Communication Systems (ICICACS), Raichur, India, 2023, pp. 1-6, doi: 10.1109/ICICACS57338.2023.10099618.
- [5] V. Jolly, M. Telrandhe, A. Kasat, A. Shitole and K. Gawande, "CNN based Deep Learning model for Deepfake Detection," 2022 2nd Asian Conference on Innovation in Technology (ASIANCON), Ravet, India, 2022, pp. 1-5, doi: 10.1109/ASIANCON55314.2022.9908862.
- [6] E. Kim and S. Cho, "Exposing Fake Faces Through Deep Neural Networks Combining Content and Trace Feature Extractors," in IEEE Access, vol. 9, pp. 123493-123503, 2021, doi: 10.1109/ACCESS.2021.3110859
- [7] A. Das, K. S. A. Viji and L. Sebastian, "A Survey on Deepfake Video Detection Techniques Using Deep Learning," 2022 Second International Conference on Next Generation Intelligent Systems (ICNGIS), Kottayam, India, 2022, pp. 1-4, doi: 10.1109/ICNGIS54955.2022.10079802.
- [8] Y. Nirkin, L. Wolf, Y. Keller and T. Hassner, "DeepFake Detection Based on Discrepancies Between Faces and Their Context," in IEEE Transactions on Pattern Analysis and Machine Intelligence, vol. 44, no. 10, pp. 6111-6121, 1 Oct. 2022, doi: 10.1109/TPAMI.2021.3093446.
- [9] D. Liu, Z. Yang, R. Zhang and J. Liu, "A Robust Deepfake Video Detection Method based on Continuous Frame Face-swapping," 2022 International Conference on Artificial Intelligence, Information Processing and Cloud Computing (AIIPCC), Kunming, China, 2022, pp. 188-191, doi: 10.1109/AIIPCC57291.2022.00048.
- [10] S. Saif, S. Tehseen, S. S. Ali, S. Kausar and A. Jameel, "Generalized Deepfake Video Detection Through Time-Distribution and Metric Learning," in IT Professional, vol. 24, no. 2, pp. 38-44, 1 March-April 2022, doi: 10.1109/MITP.2022.3168351.
- [11] Ranjan, Sarvesh Patil, Faruk Kazi, "Improved Generalizability of Deep-Fakes Detection Using Transfer Learning Based CNN Framework", (IEEE 2020).
- [12] L. Rebello, L. Tuscano, Y. Shah, A. Solomon and V. Shrivastava, "Detection of Deepfake Video using Deep Learning and MesoNet," 2023 8th International Conference on Communication and Electronics Systems (ICCES), Coimbatore, India, 2023, pp. 1022-1026, doi: 10.1109/ICCES57224.2023.10192854.
- [13] N. S. Ivanov, A. V. Arzhskov and V. G. Ivanenko, "Combining Deep Learning and Super-Resolution Algorithms for Deep Fake Detection," 2020 IEEE Conference of Russian Young Researchers in Electrical and Electronic Engineering (EIConRus), St. Petersburg and Moscow, Russia, 2020, pp. 326-328, doi: 10.1109/EIConRus49466.2020.9039498.
- [14] T. Jung, S. Kim and K. Kim, "DeepVision: Deepfakes Detection Using Human Eye Blinking Pattern," in IEEE Access, vol. 8, pp. 83144-83154, 2020, doi: 10.1109/ACCESS.2020.2988660.
- [15] K. Jalui, A. Jagtap, S. Sharma, G. Mary, R. Fernandes and M. Kolhekar, "Synthetic Content Detection in Deepfake Video using Deep Learning," 2022 IEEE 3rd Global Conference for Advancement in Technology (GCAT), Bangalore, India, 2022, pp. 01-05, doi: 10.1109/GCAT55367.2022.9972081.